

Annotation Protocol of **RiceLeafDiseaseBD**: A high-resolution annotated smartphone image dataset of healthy and diseased rice leaves from Bangladesh

Dataset Characteristics

Table 1: Dataset Statistics Overview

Statistic	Value	Details
Total Images	9,769	6 classes
Total Bounding Boxes	74,836	Region-level annotations
Average Boxes per Image	~ 8	Varies by disease class
Image Resolution	1024×1024	Standardized preprocessing

1 Purpose

This document describes the standardized protocol developed for creating region-level annotations of diseased regions on rice leaves (*Oryza sativa*). The protocol aims to minimize subjectivity and variability in the annotation task, ensuring consistent and reproducible results across all annotators.

Protocol Objectives:

1. Standardized disease identification
2. Consistent bounding box placement
3. Expert-validated annotations
4. YOLO-compatible format output

Figure 1: Primary objectives of the annotation protocol

2 Scope

Region-level annotations are provided for images belonging to the following five disease classes:

Table 2: Disease Classes and Causal Agents

Disease Class	Causal Agent/Description
Blast	Caused by <i>Magnaporthe oryzae</i>
Brown Spot	Caused by <i>Bipolaris oryzae</i>
Leaf Smut	Caused by <i>Entyloma oryzae</i>
Rice Tungro	Viral disease transmitted by leafhoppers
Sheath Blight	Caused by <i>Rhizoctonia solani</i>
Healthy	No visible disease symptoms (empty annotations)

3 Annotation Strategy

Manual annotation challenges and solutions:

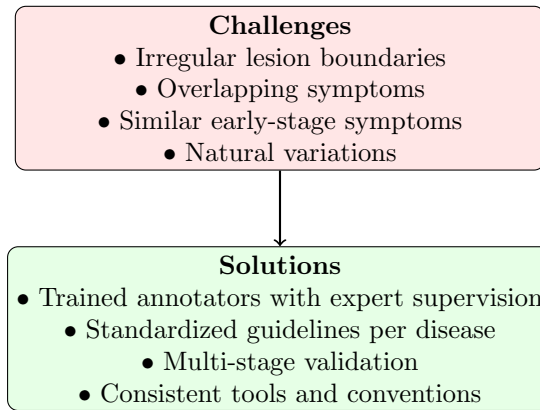


Figure 2: Annotation challenges and corresponding solutions

6 Annotation Guidelines by Disease Class

Table 3: Disease-Specific Annotation Guidelines

Disease	Annotation Rules
Blast	• Spindle-shaped lesions with gray-white centers • Brown margins • Individual lesions marked separately • Size: 1–2 cm length
Brown Spot	• Circular to oval spots • Brown centers • Each distinct spot individually • High density possible
Leaf Smut	• Small black spots/streaks • Follow leaf veins • All visible pustules included • Size: 2–5 mm
Rice Tungro	• Yellowing/discoloration patterns • Stunted growth areas • Mottled chlorotic patterns • Diffuse symptoms
Sheath Blight	• Large, irregular lesions • Greenish-gray to brown • Water-soaked borders • Size: 1–3 cm or larger

9 Conversion to YOLO Format

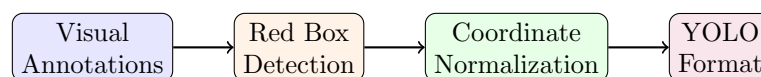


Figure 3: Conversion pipeline from visual annotations to YOLO format

9.3 Class ID Mapping

Table 4: YOLO Format Class ID Mapping

Class ID	Disease Class	Sample Count	Avg. Boxes/Image	Total Boxes
0	Brown Spot	2,178	~10	21,885
1	Blast	1,326	~9	12,279
2	Healthy	1,575	-	-
3	Leaf Smut	724	~12	8654
4	Rice Tungro	2,244	~7	14,762
5	Sheath Blight	1,722	~10	17,256

Listing 1: YOLO Format Example

```
# Example annotation file: brown_spot_001.txt
0 0.512695 0.651855 0.005859 0.006836
0 0.602051 0.639648 0.004883 0.005859
0 0.473145 0.839355 0.051758 0.083008

# Format: <class_id> <center_x> <center_y> <width> <height>
# All coordinates normalized to [0, 1] range
```

11 Quality Control Pipeline

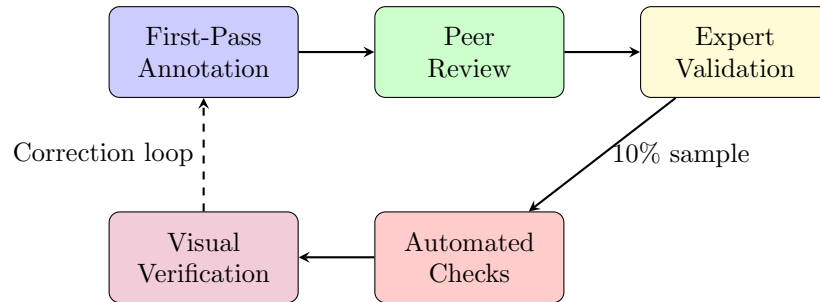


Figure 4: Quality control workflow with multi-stage validation

12 Inter-Annotator Agreement Results

Inter-Annotator Agreement Metrics (n=200 images)

Metric	Value	Interpretation	Visualization
Cohen's κ	0.82	Substantial agreement	82%
IoU Score	0.78	Good overlap	78%
F1-Score	0.85	High detection consistency	85%

Table 5: Inter-annotator agreement assessment results

15 Intended Use Cases

Table 6: Potential Applications and Use Cases

Application Area	Specific Use Cases
Computer Vision Research	• Benchmarking (YOLO, Faster R-CNN) • Transfer learning studies • Multi-task learning
Agricultural Applications	• Mobile diagnostic apps • Precision agriculture • Disease severity assessment
Data Science	• Lesion counting algorithms • Pattern recognition • Anomaly detection

16 Reproducibility Framework

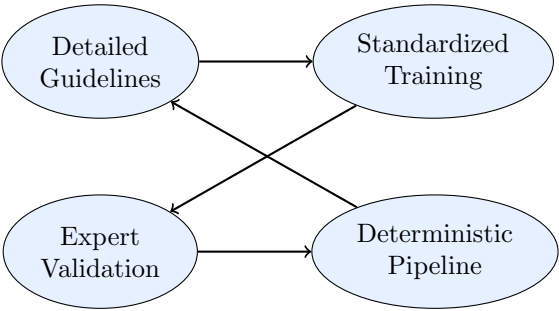


Figure 5: Reproducibility framework ensuring consistent annotations

17 Ethical Considerations

Ethical Framework

- Farm owner permissions obtained
- Respect for local agricultural practices
- Farmer expertise acknowledged and valued
- Dataset publicly available for agricultural research
- Commitment to ethical AI in agriculture

End of Annotation Protocol